ASE CHAPTER 7

T4 Visual Models & Diagram Interpretation Pack

Q7.V01 · Hardware Bathtub Curve and Software Reliability Curve (Exercise 7.2)

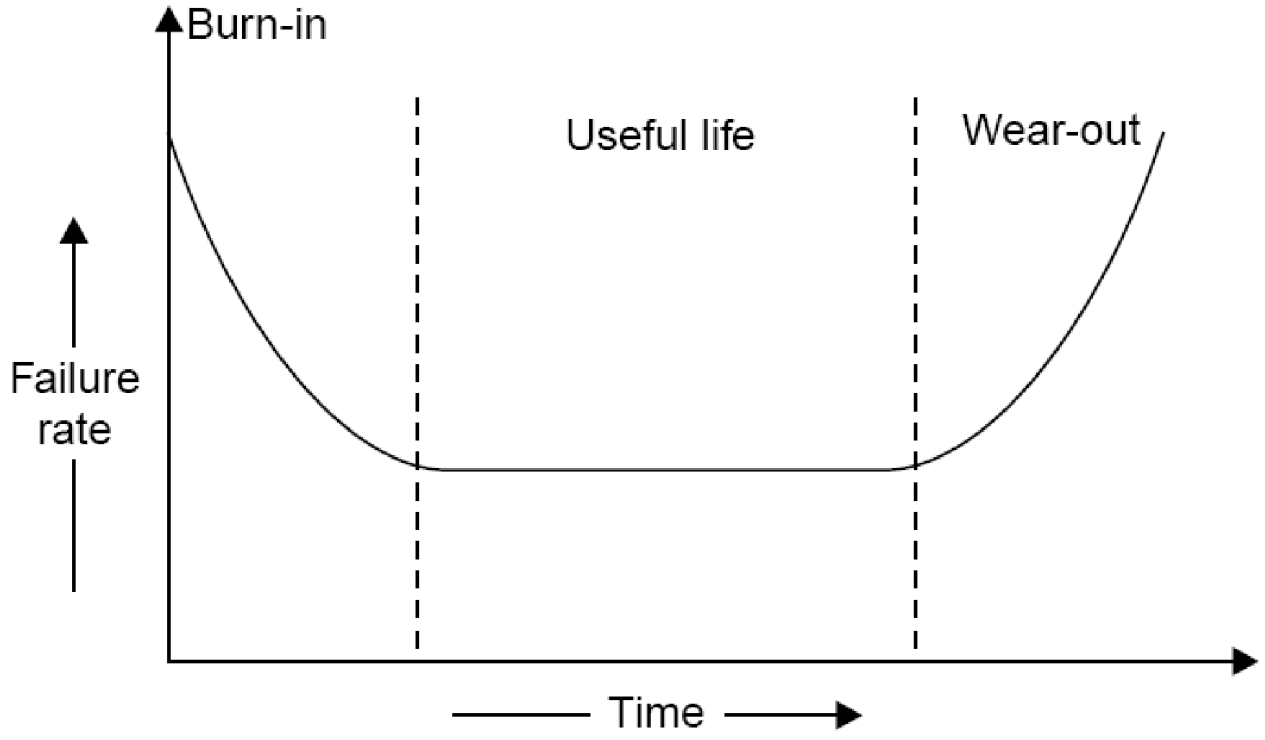


Fig. 7.1: Bath tub curve of hardware reliability.

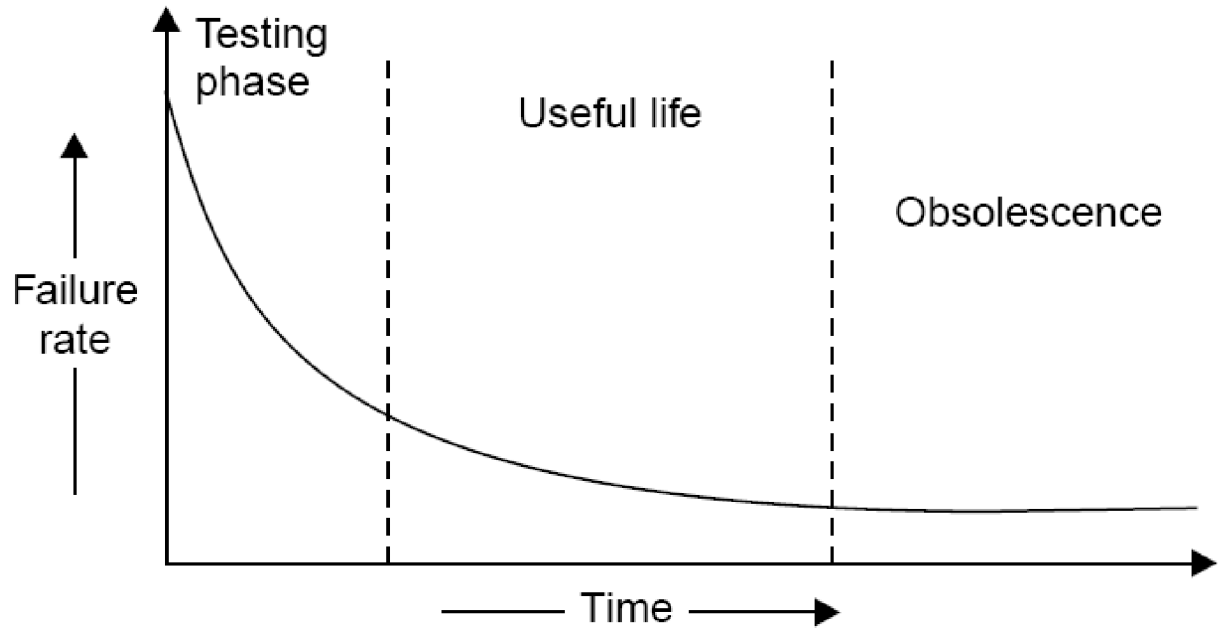


Fig. 7.2: Software reliability curve (failure rate versus time)

1. **Figure 7.1:** Hardware failure rate falls during burn-in, stays nearly constant through useful life, then rises during wear-out.
2. **Figure 7.2:** Software failure rate falls during testing, remains low through useful life, and reaches the obsolescence phase. This plotted curve has no hardware-style wear-out phase.

Academic Note

ملاحظة المخطط: يوضح الشكل الأول ارتفاع فشل العتاد بعد مرحلة العمر النافع، بينما يوضح الثاني انخفاض فشل البرمجيات أثناء الاختبار واستقراره خلال العمر النافع ثم تقادمها.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.1.1, pp. 308–310; companion slides 3–4, Figs. 7.1–7.2.

Q7.V02 · Input Space and Example Input-State Probabilities (Exercise 7.6)

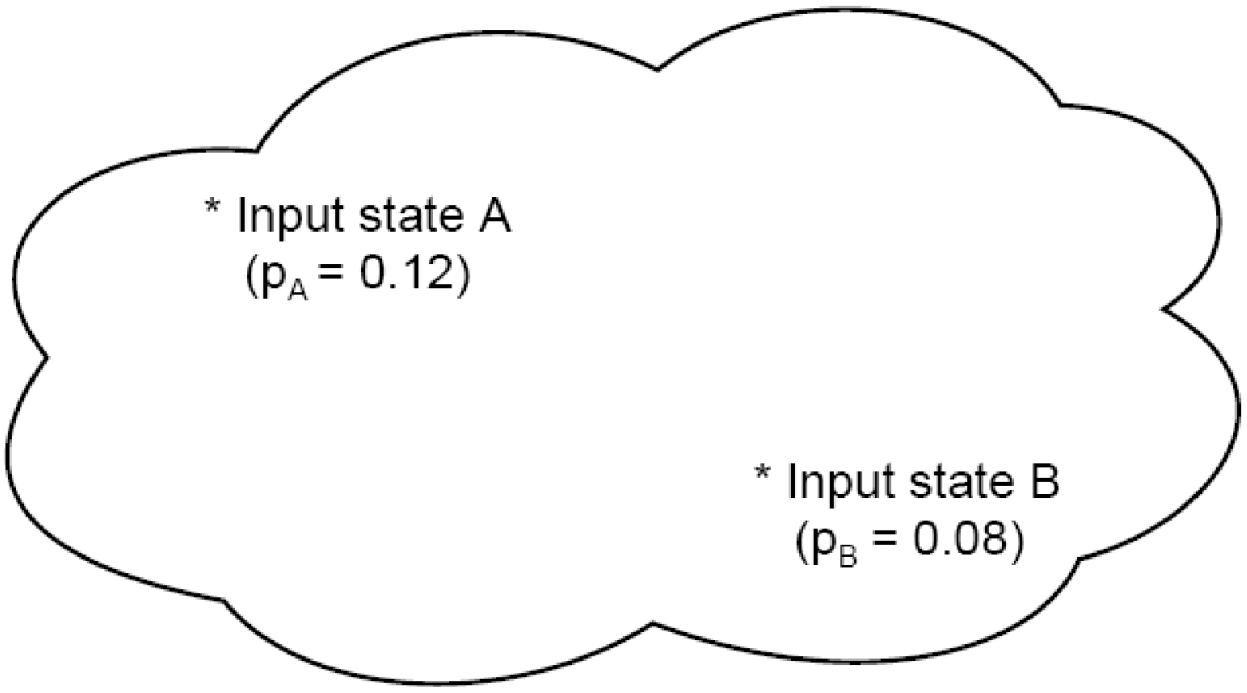


Fig. 7.4: Input Space

- The cloud labelled **Input Space** shows two example states: A with probability 0.12 and B with probability 0.08.
- These labels associate occurrence probabilities with the two states; the figure does not mark a failure domain.

Academic Note

ملاحظة المخطط: الشكل يعرض حالتي إدخال واحتماليهما؛ لا يرسم منطقة فشل.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.1.4, pp. 314–316; companion slide 18, Fig. 7.4.

Q7.V03 · Probability Portions in an Operational Profile (Exercise 7.6)

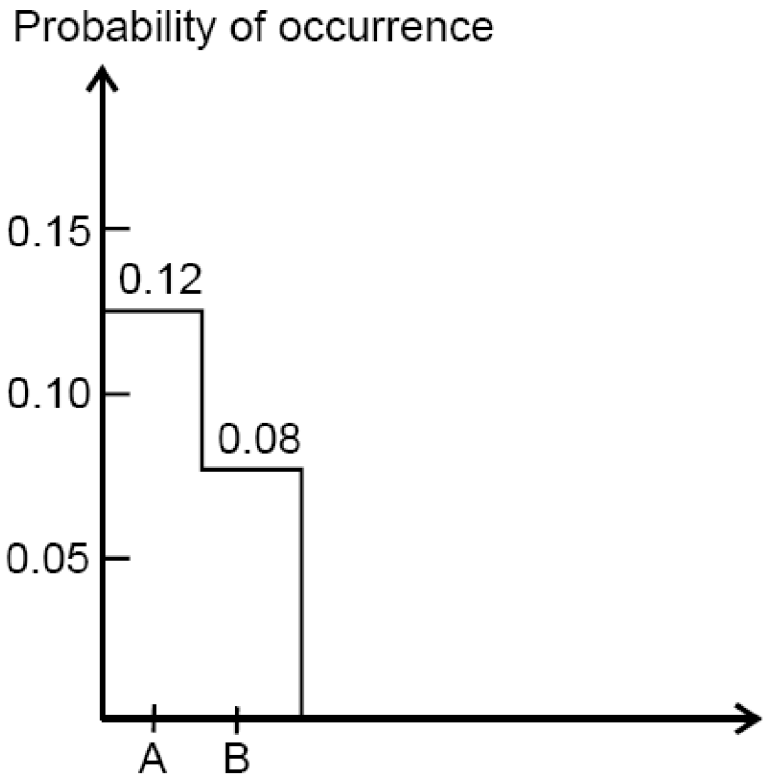


Fig. 7.5: Portion of operational profile

- The vertical axis is probability of occurrence; the input states A and B are shown with probabilities 0.12 and 0.08.
- The caption identifies this as a portion of an operational profile, not a complete branching tree.

Academic Note

ملاحظة المخطط: يبين الجزء المعروض احتمال ظهور الحالة A والحالة B؛ لا يعرض شجرة تشغيلية.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.1.5, pp. 316–318; companion slide 19, Fig. 7.5.

Q7.V04 · McCall Quality Factors and Criteria (Exercises 7.10 & 7.11)

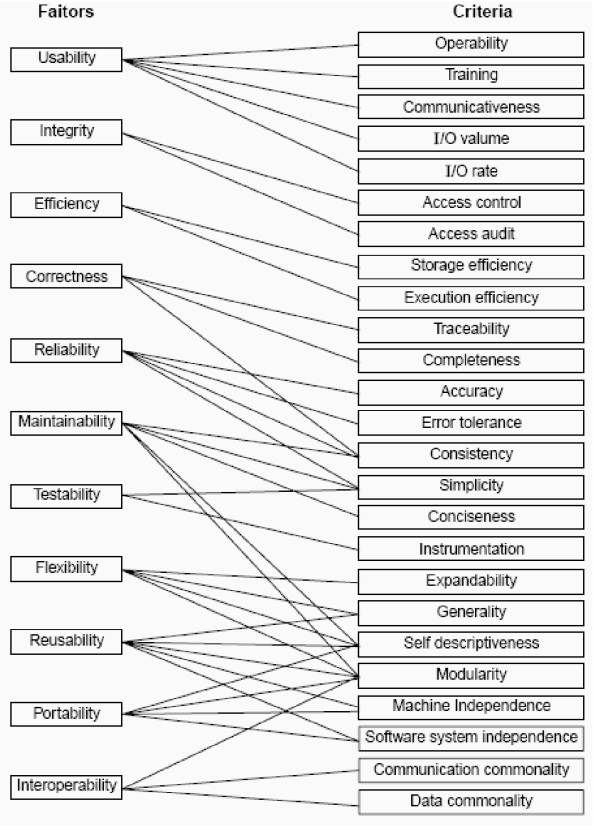


Fig 7.10: McCall's quality model

- The figure maps 11 quality factors to 25 quality criteria. Its connecting lines show which criteria are associated with each factor; a criterion can connect to more than one factor.
- This is a factor-to-criteria map; it does not show the separate Product Operation, Product Revision, and Product Transition grouping.

Academic Note

ملاحظة المخطط: يربط الشكل 11 عاملاً للجودة بـ25 معياراً؛ تجميع Operation وRevision وTransition يخص شكلاً آخر.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.2.1, p. 322; companion slide 35, Fig. 7.10.

Q7.V05 · Boehm Software Quality Model (Exercise 7.12)

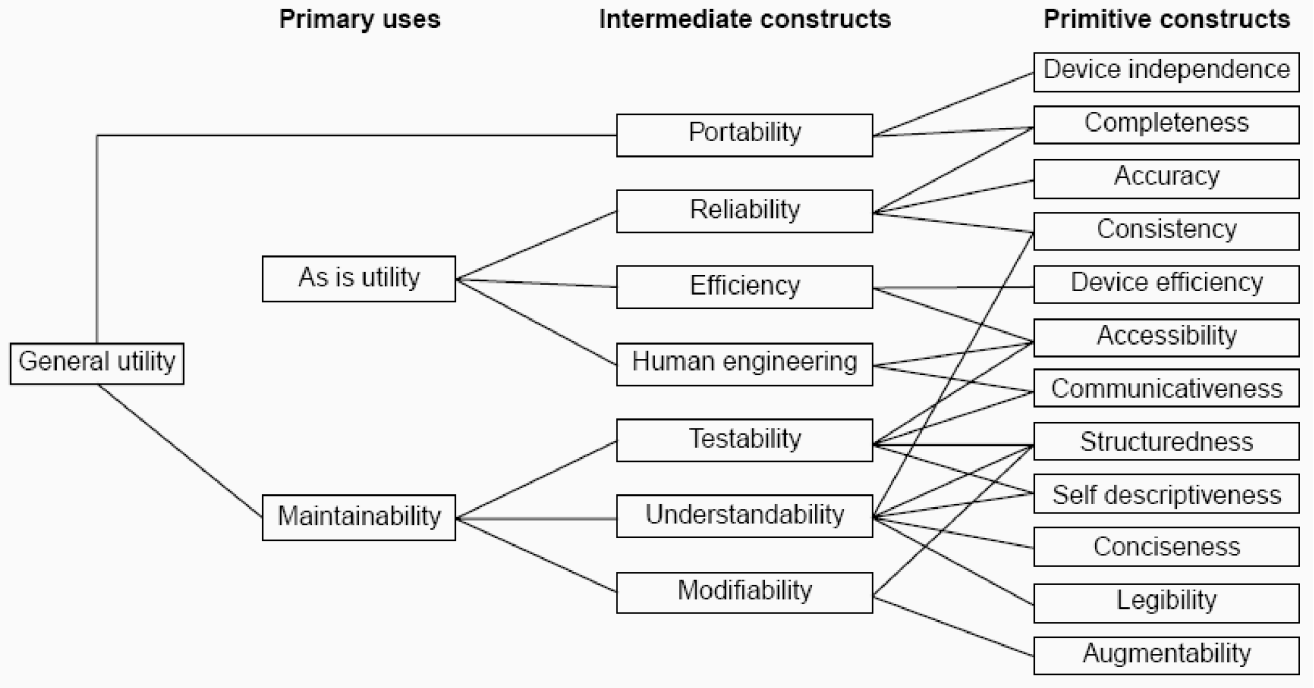


Fig.7.11: The Boehm software quality model

- The figure shows three levels: primary uses, intermediate constructs, and primitive constructs.
- General Utility links directly to Portability and Maintainability. As-is Utility links to Reliability, Efficiency, and Human Engineering; Maintainability links to Testability, Understandability, and Modifiability.
- The figure contains 7 intermediate constructs and 12 primitive constructs.

Academic Note

ملاحظة المخطط: يتدرج نموذج Boehm من الاستخدامات إلى التراكيب الوسيطة ثم الخصائص الأولية؛ يضم الشكل 7 تراكيب وسيطة و12 خاصية أولية.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.2.2, pp. 324–325; companion slide 40, Fig. 7.11.

Q7.V06 · ISO 9126 Quality Model (Exercise 7.13)

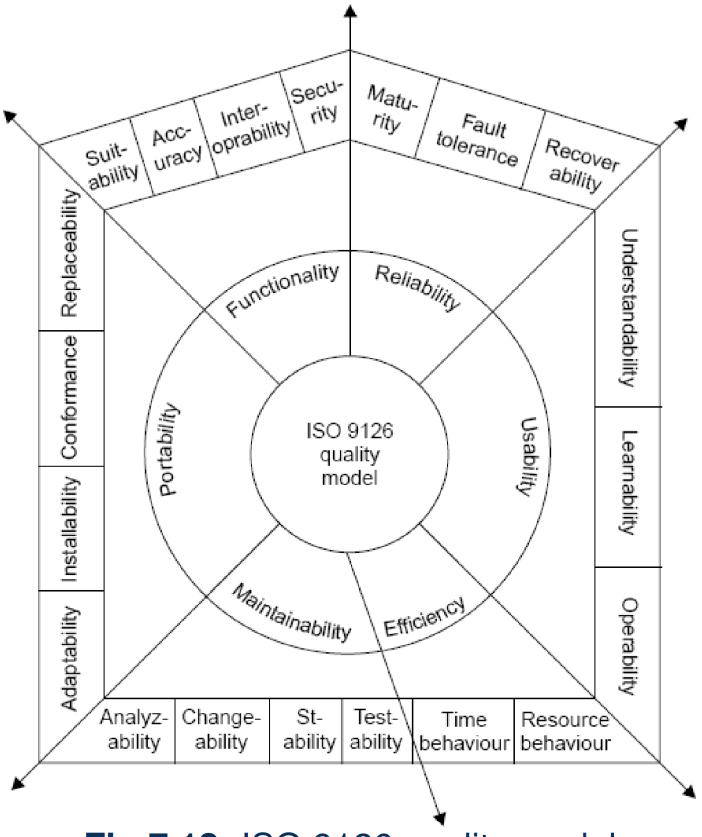


Fig.7.12: ISO 9126 quality model

The figure groups 20 subcharacteristics under six characteristics:

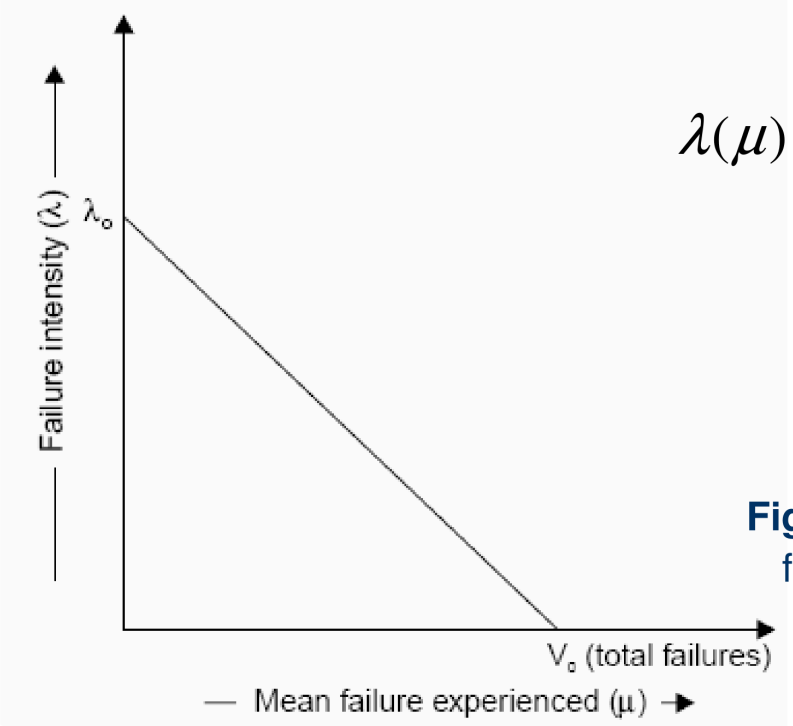
1. **Functionality:** Suitability, Accuracy, Interoperability, Security.
2. **Reliability:** Maturity, Fault tolerance, Recoverability.
3. **Usability:** Understandability, Learnability, Operability.
4. **Efficiency:** Time behaviour, Resource behaviour.
5. **Maintainability:** Analyzability, Changeability, Stability, Testability.
6. **Portability:** Adaptability, Installability, Conformance, Replaceability.

Academic Note

ملاحظة المخطط: الشكل يعرض 6 خصائص رئيسية و20 فرعية؛ Conformance تابعة لـPortability.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.2.3, pp. 326–328; companion slide 46, Fig. 7.12.

Q7.V07 · Basic and Logarithmic Poisson Failure-Intensity Curves (Exercises 7.15 & 7.17)



$$\lambda(\mu) = \lambda_0\left(1 - \frac{\mu}{V_0}\right) \quad (1)$$

Fig.7.13: Failure intensity λ as a function of μ for basic model

$$\lambda(\mu) = \lambda_0 \exp(-\theta\mu)$$

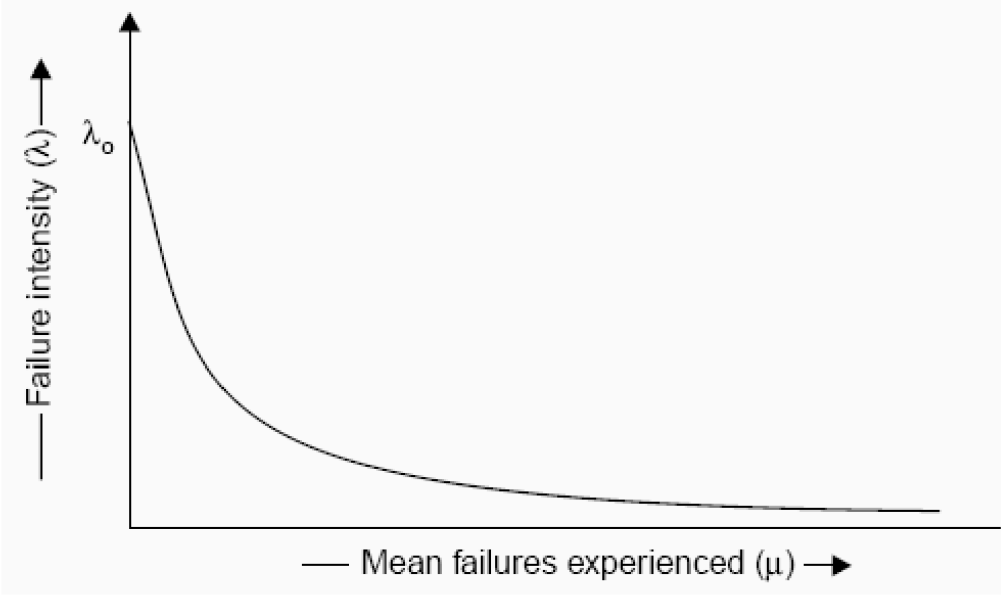


Fig.7.18: Relationship between μ & λ

1. **Basic model, Figure 7.13:** Plots failure intensity λ against mean failures experienced μ. The straight line falls from λ_0 at $\mu = 0$ to zero at $\mu = V_0$; the displayed equation is $\lambda(\mu) = \lambda_0(1 - \mu/V_0)$.
2. **Logarithmic Poisson model, Figure 7.18:** Also plots λ against μ, using $\lambda(\mu) = \lambda_0 \exp(-\theta\mu)$. Its curve falls steeply at first, then gradually flattens toward zero.

> 📝 **Academic Note**
>
> **ملاحظة المخطط:** الشكلان كلاهما λ مقابل μ: الأساسي خط مستقيم، واللوغاريتمي منحنى يتناقص ثم يتسطح.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.3.1, p. 330, and §7.3.2, p. 336; companion slides 47 and 59, Figs. 7.13 and 7.18.

Q7.V08 · Calendar-Time to Execution-Time Ratio and Limiting Resources (Exercise 7.23)

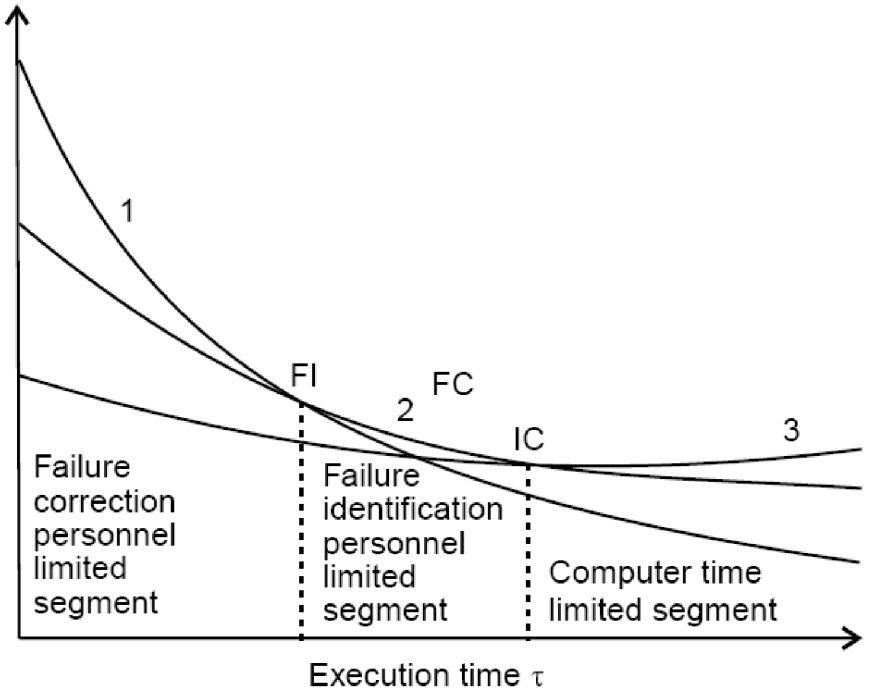


Fig.7.21: Calendar time to execution time ratio for different limiting resources

- The plot relates the calendar-time/execution-time ratio to execution time τ.
- It labels three resource-limited segments: failure-correction personnel, failure-identification personnel, and computer time.

> **Academic Note**
> **ملاحظة المخطط:** القطاعات الثلاثة المعلّمة هي محدودية موظفي التصحيح، ثم موظفي تحديد الفشل، ثم وقت الحاسوب.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.3.3, pp. 340–342; companion slide 77, Fig. 7.21.

Q7.V09 · Jelinski–Moranda Failure-Intensity Steps (Exercise 7.25)

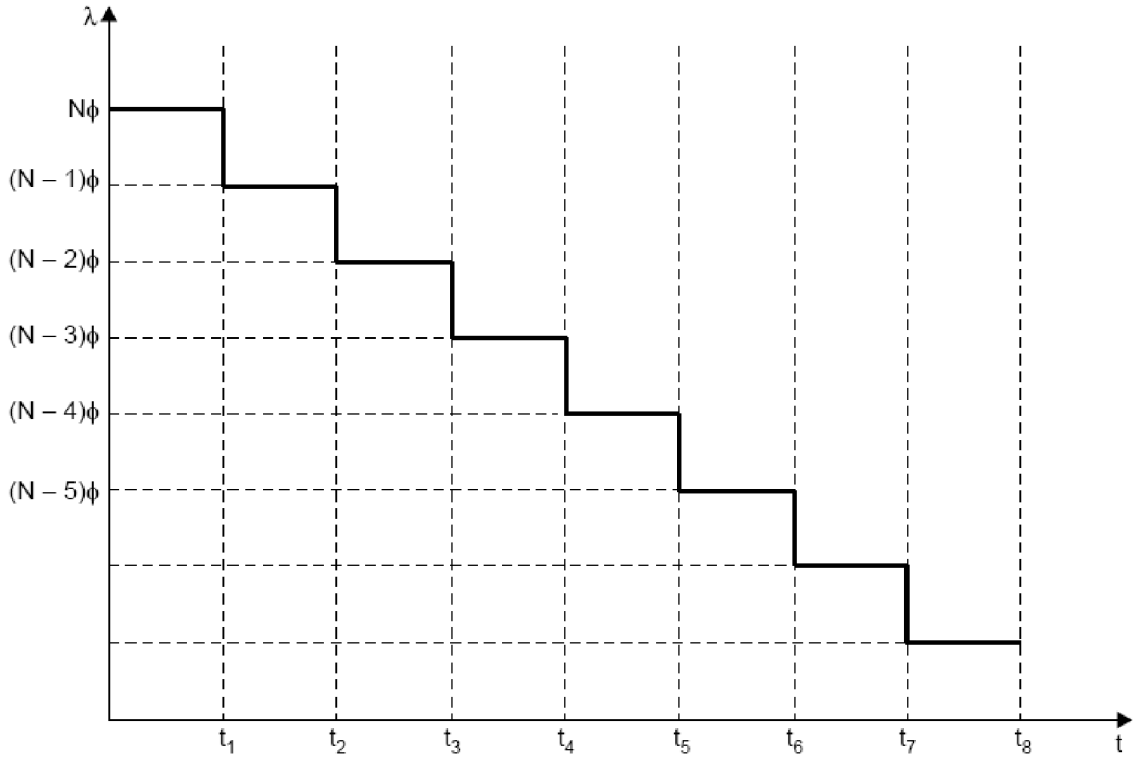


Fig.7.22: Relation between t & λ

- The plot shows failure intensity λ against time t. The rate remains constant between failures and drops by φ at successive failure times, from Nφ to (N − 1)φ and onward.

> **Academic Note**
> **ملاحظة المخطط:** يبقى λ ثابتاً بين أزمنة الفشل، ثم ينخفض بمقدار φ عند كل خطوة لاحقة.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.3.4, pp. 343–345; companion slide 90, Fig. 7.22.

Q7.V10 · SEI CMM Maturity Levels and Characterizations (Exercises 7.27 & 7.28)

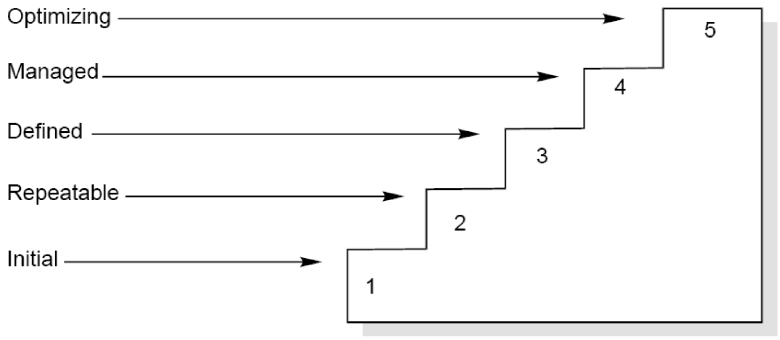


Fig.7.23: Maturity levels of CMM

Maturity Level	Characterization
Initial	Adhoc Process
Repeatable	Basic Project Management
Defined	Process Definition
Managed	Process Measurement
Optimizing	Process Control

Fig.7.24: The five levels of CMM

- Figure 7.23 orders the five levels: Initial (1), Repeatable (2), Defined (3), Managed (4), and Optimizing (5).
- Figure 7.24 pairs them with these characterizations: Adhoc Process, Basic Project Management, Process Definition, Process Measurement, and Process Control, respectively.

Academic Note

ملاحظة المخطط: الشكل الأول يبيّن ترتيب المستويات الخمسة، والثاني يربط كل مستوى بوصفه المقابل.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §7.4.1, pp. 347–350; companion slides 94 and 96, Figs. 7.23–7.24.